GRAVITYSORT · PROJECT PLAN

# GravitySort project plan

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## DOCUMENT CONTROL

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| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/gravitysort | | |
| Source | docs/11-plan.md @ 6c7e5bc | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.1 | 2026-10-25 | Amish Chadha | First issue (kit 1.8 pipeline): workstreams, resources, stage gates P0 to P6 (P4 to P6 on hold), next steps, decisions requested and the reference list |

# GravitySort project plan

GravitySort is at stage P3 (design outputs, TRL 3): the constructable design, its D3 model of every concept option, drawings, priced BOM, build plan and pipeline documents exist on paper. The next gate, P4 (verification, TRL 4), is on hold under the portfolio cap; before it Amish is asked to confirm the concept (option A recommended, GVS-DEC-001 O1) and to decide the containment band (O2). Method: STANDARDS sections 19 and 23.

## Workstreams

Table 1. Workstreams.

| Workstream | Owner (role) | Deliverable | Status |
|---|---|---|---|
| Concept and STEM | Design lead | GVS-STM-001, GVS-OPT-001, GVS-DDR-004 | Done at P3; STEM check open: nine rows need tests or datasheets, two need Amish's decisions (O2, O3) |
| Requirements and traceability | Design lead | GVS-URS-001, GVS-PRS-001, GVS-RTM-001 | Done at P3; to revise after co-design |
| Calculations | Design lead | GVS-CAL-001 v0.9 | Done at P3 |
| CAD and drawings | Design lead | `cad/src/model.py` (D3, options A to C), STEP, GVS-DWG-001 to 121 | Done at P3 |
| Build plan and BOM | Design lead | GVS-BLD-001, `bom/bom.csv` | Done at P3 |
| Regulatory, risk and market | Project lead | GVS-REG-001, GVS-RSK-001, GVS-MKT-001 | First issue; items to assess with the partner |
| Partner and co-design | Project lead (Amish) | Agreement with the first candidate cooperative; co-design sessions | Not started (outreach is Amish's action) |
| Renders and media | Coordinator | Options picture, product renders, cards (Blender on Amish's Mac) | Scenes ready; renders pending |
| Prototype build and lab tests (P4) | Workshop partner and design lead | Prototype, test plans and reports | On hold (TRL cap 3) |
| Field trial (P5) | Partner organization | Trial with the cooperative | On hold |

## Resources needed

Table 2. Resources (estimates, to assess with the partner before P4).

| Resource | Deliverable | Effort | Estimated cost (USD) |
|---|---|---|---|
| Prototype parts | One machine to the BOM | n/a | 594 (BOM estimate) plus 335 if the MotionCore option is built |
| Workshop time (welder, fitter) | Frame, guards, table, assembly | To assess with the workshop | To assess |
| PU casting trial | Liner cast in the printed mold | To assess | To assess |
| Spiked or characterized test ore and assays | Recovery test of PR-004 | To assess | To assess |
| Partner co-design sessions | Validated user requirements | To assess | To assess |

## Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

| Stage | Gate criteria | Status |
|---|---|---|
| P0 Proposal | PRB, STM, OPT, REG, FEA, PLN | Met (2026-10-25, retrofitted under kit 1.8) |
| P1 Definition | URS, PRS draft, RSK with hazard analysis, MKT, REG complete | Met (2026-10-25) |
| P2 Design inputs | PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed where a calculation can close it | Met on paper; concept selection awaiting Amish (GVS-DDR-004) |
| P3 Design outputs | Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability complete | Met (TRL 3 since 2026-09-25; D3 model of every option 2026-10-25) |
| P4 Verification | Prototype, lab test plans and reports (recovery, table ratio, pedal power, containment impact test) | On hold (TRL cap 3) |
| P5 Validation | Field trial with the partner, human factors | On hold |
| P6 Commercialization | Only with a named deployment partner | On hold |

## Next steps and decision requested

1. **Decision O1 (concept selection, GVS-DDR-004):** confirm option A, the current design, as the concept carried forward (recommended), or choose B or C.
2. **Decision O2 (containment):** keep the 6 mm tub with an impact test before any spin test (recommended), or fit a steel band now, which takes the mass over PR-011 (GVS-DEC-001).
3. **Decisions O3 to O5:** tank sway near the table frequency (raise the brace node and take a saving, recommended), spindle supports (tap test at TRL 4), fluidization flow (sweep it in the TRL 4 test).
4. Amish's outreach to the first candidate partner (decided 2026-10-02), so that co-design can test the user requirements.

5. Coordinator renders `media/options.png` and the product renders on Amish's Mac.
6. When the TRL cap lifts: the P4 tests in Table 3, starting with the containment impact test and the recovery test.

## References

Every source cited in this repository's documents, Chicago style.


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